

Galois Covers of M_{23} over $\mathbb{Q}$: A Positive-Dimensional Approach

Braid-Orbit Reconstruction, Exact Clutching, Algebraic Collision Tails, and Six-Branch Refinements

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Revised: August 19, 2026 (June 23, 2026)

Abstract

For $M_{23} \leq S_{23}$, the ordered passport $(2A, 2A, 3A, 5A)$ has exactly 980 generating inner Nielsen classes in one pure-braid orbit and gives a genus-119 cross-ratio cover. The all-orderings orbit has 11,760 states; a free Klein-four quotient gives a degree-2940, genus-43 reduced j -line cover with 406 cusps. The real-boundary computation certifies eight individual $\mathbb{Q}$ -rational boundary points, and $B_{27} = \mathbb{Q}[Z]/(Z^{27} - Z)$ supplies the target normal form for the interior fiber-algebra criterion.

For the five-branch passport $(2A, 2A, 2A, 2A, 3A)$, an independent computation gives a connected 21,456-state colored component. All six equal- $2A$ collision divisors have 980 width-five cycles, and admissible-cover clutching identifies every normalized width-five component with the complete genus-119 four-branch curve. The released $(2A, 23A, 23B)$ tuple admits a generating five-branch refinement. A common six-branch class $(2A^3, 3A^3)$ joins the two five-branch components through an exact 14-move colored braid word and rational collision tails. A generating $2A^{10}$ refinement has source genus 18; an exact 29,005-move B_{10} word connects its $2 + 4 + 4$ contraction to the released three-point tuple, while a separate h -equivariant ten-involution refinement fuses by five pairs into the verified five-branch component.

The mixed collision boundary is a degree-18, genus-2 cover. On the canonical ten-state cell, the central deck involution gives the phase pairing commuting with q_1 , while the native return conjugates q_1 to q_1^{-1} and yields a dihedral order-10 action. The associated set-stabilizer base change produces a degree-10 Hurwitz subcover, a degree-5 quotient of genus 161, and the comparison morphism $\mathfrak{B}_H$. The carrier $\beta_{10}(t) = 1 - (1 - t^2)^5$ quotients to $\beta_5(t) = 1 - (1 - t)^5$. The width-11 locus consists of four transitive degree-66, genus-zero components paired by an explicit inverse-return involution. Pair-clutching and balanced twisted-cover deformation complete the corresponding $\mathbb{Q}$ -defined admissible-cover geometry.

Keywords. Mathieu group M_{23} ; Hurwitz spaces; Nielsen classes; braid orbits; admissible covers; inverse Galois theory; computational group theory.

1 Introduction

The inverse Galois problem asks whether a finite group occurs as the Galois group of a finite extension of $\mathbb{Q}$, and more strongly whether it occurs regularly over $\mathbb{Q}(t)$. Huang, Jackson, Lee, Poonen, Pries, and Zhang proved the case of M_{23} through the non-rigid three-point class triple

$$(2A, 23A, 23B).$$

Their Nielsen class has seven elements, the associated degree-23 quotient curve has genus 4, and their reconstruction yields an explicit regular M_{23} -extension over $\mathbb{Q}(t)$ together with explicit degree-23 specializations over $\mathbb{Q}$ [4].

The class vector

$$\mathbf{C}_4 = (2A, 2A, 3A, 5A)$$

defines a positive-dimensional Hurwitz problem with a degree-980, genus-119 ordered component. The five-branch vector

$$\mathbf{C}_5 = (2A, 2A, 2A, 2A, 3A)$$

defines a 21,456-state colored component whose width-five boundary returns the four-branch curve. The explicit three-point tuple also admits a five-branch refinement of type $(2A, 3A, 3A, 3A, 3A)$. A

six-branch common refinement and a ten-involution refinement connect these constructions through exact braid and clutching maps.

Häfner records the numerical invariants 980 and 119 for the four-branch component [3]. The computations below independently reconstruct the complete four-branch generating Nielsen class, determine the ordered and reduced permutation geometry, reconstruct the 21,456-state five-branch component, prove the full six-direction width-five boundary by exact clutching and equal-class braid symmetry, construct the six- and ten-branch common refinements, and give explicit rational models for all collision tails used in those refinements.

The all-orderings orbit has 11,760 states, and its free Klein-four quotient has degree 2940, genus 43, and 406 cusps. The real-boundary computation identifies eight individual $\mathbb{Q}$ -rational boundary points, while the degree-27 algebra B_{27} supplies the target normal form for the interior fiber-algebra criterion. At the five-branch boundary, the exact finite calculations also isolate the degree-18, genus-2 mixed carrier, the canonical ten-state stabilizer pullback, and the four degree-66, genus-zero width-11 components.

For broader inverse-Galois background and braid methods, see [8]; for explicit Hurwitz-space computation in the Mathieu setting, see [7]. Fields of definition of Hurwitz components are treated in [9]. The admissible-cover compactification originates in [6] and is used below through the balanced twisted-cover and Hurwitz-space frameworks of [5, 10].

The cyclic degree-five cover

$$\beta_5(t) = 1 - (1 - t)^5$$

contributes a finite étale fixed/dual structure. Its zero fiber contains a canonical rational direct factor and a complementary cyclotomic factor with quadratic inversion quotient. The canonical ten-state cell has two exact involutions with distinct roles. The phase pairing

$$\kappa_{\text{pair}}(n, p) = (n, p + 1)$$

commutes with $q_1(n, p) = (n + 1, p)$, agrees with the restriction of the central deck involution after the canonical set-stabilizer base change, and gives the algebraic model $\beta_{10}(t) = 1 - (1 - t^2)^5 \rightarrow \beta_5$. The native braid return

$$\kappa_{\text{ret}}(n, p) = (1 - n, p + 1)$$

conjugates q_1 to q_1^{-1} and supplies the dihedral ten-state return action. The mixed boundary has degree 18 and genus 2; its width-11 locus consists of four degree-66, genus-zero components paired by an explicit inverse-return involution. The ten-state set-stabilizer pullback, pair-clutching, and balanced-node deformation theory supply the corresponding global boundary and admissible-cover geometry.

Chronology and independence. The four-branch reconstruction, reduced j -line computation, boundary analysis, five-branch component, clutching calculations, and coefficient systems were developed independently of the construction of Huang, Jackson, Lee, Poonen, Pries, and Zhang. Their public announcement on August 9, 2026 [4] prompted comparison with the released $(2A, 23A, 23B)$ tuple. The resulting five-branch refinement, six-branch comparison bridge, and ten-involution comparison layer are identified explicitly in section 7; each takes the published tuple of [4] as input. The present article is offered as a complementary study of the positive-dimensional Hurwitz geometry.

1.1 Main verified result

Theorem 1.1 (Verified Hurwitz geometry). *The following statements hold.*

- (i) *The ordered inner and absolute Nielsen classes of type $(2A, 2A, 3A, 5A)$ coincide, contain exactly 980 generating classes, and form one pure-braid orbit.*

- (ii) *The ordered cross-ratio cover has degree 980 and genus 119. Its boundary passport gives a $\mathbb{Q}$ -rational divisor of degree one, hence index and period one. The real-boundary verification certifies eight individual $\mathbb{Q}$ -rational boundary points, while the width-one degree-two divisors over $\lambda = 1$ and $\lambda = \infty$ are imaginary quadratic.*
- (iii) *The full all-orderings braid orbit has 11,760 states. A free Klein-four quotient has degree 2940, genus 43, and 406 cusps. Its reduced elliptic monodromy over $j = 1728$ is fixed-point-free of type 2^{1470} .*
- (iv) *The colored five-branch component of type $(2A, 2A, 2A, 2A, 3A)$ has 21,456 states. All six collision divisors among the four equal $2A$ labels carry 980 width-five cycles, and clutching gives a bijection from each width-five stratum to the complete four-branch Nielsen class.*
- (v) *The four- and five-branch ramification constraints admit exact universal coefficient systems with respectively 48 and 72 coefficient equations. The 27-dimensional finite étale algebra $B_{27} = \mathbb{Q}[Z]/(Z^{27} - Z)$ is an exact target normal form for interior-fiber activation, with three rational rank-one trace-one projectors. Component selection and source activation use the Nielsen data and a genuine degree-980 interior fiber algebra.*
- (vi) *The released three-point tuple of type $(2A, 23A, 23B)$ has an exact generating five-branch refinement of type $(2A, 3A, 3A, 3A, 3A)$. For either printed 23-element, there are exactly 138 ordered $3A \cdot 3A$ factorizations; the ordered types $2A \cdot 2A$, $2A \cdot 3A$, and $3A \cdot 2A$ each have factorization count zero.*
- (vii) *The two five-branch components admit a common six-branch refinement of type $(2A^3, 3A^3)$. Within refinements using local classes in $\{2A, 3A\}$ and one elementary fusion on each side, six branches are minimal. The two specified collision faces land in the verified five-branch component and in the five-branch component containing the refinement of the released three-point tuple, respectively.*
- (viii) *Both collision faces admit explicit $\mathbb{Q}$ -defined genus-zero tail models whose restricted passports agree exactly with the degree-23 permutation data.*
- (ix) *On the mixed boundary with normal inertia q_4^2 , the fixed fiber has 18 states and boundary monodromy group of order 92,897,280. Its unique nontrivial central deck involution acts freely on the 18-sheet geometric fiber F_{45} . The ten-state subset R_{10} has an orbit of 126 subsets; its set stabilizer has order 737,280 and acts transitively on R_{10} through a group of order 3840. The corresponding degree-126 finite étale base change produces a connected degree-10 Hurwitz subcover. The central deck involution restricts exactly to κ_{pair} on R_{10} , and the connected degree-5 quotient has induced group S_5 . Quotient descent gives the comparison morphism $\mathfrak{B}_H$ of theorem 6.9. The compactified base-change curve, ten-sheet subcover, and five-sheet quotient have genera 31, 327, and 161, respectively.*
- (x) *The target five-branch seed admits a product-one generating $2A^{10}$ refinement of source genus 18. A 29,005-move B_{10} word reaches a ten-tuple whose $2 + 4 + 4$ contraction is simultaneously conjugate to the released three-point tuple. Separately, the released tuple admits a ten-involution refinement equivariant under the explicit branch-cycle involution induced by h , and its five-pair fusion lies in the verified target five-branch component.*
- (xi) *A second h -equivariant refinement, obtained by nine local B_4 moves, fuses to a generating $(2A, 2A, 3A, 2A, 2A)$ tuple in the verified target component by a nine-move B_5 word. One refined leaf a has $\text{ord}(ha) = 5$; with $r = ha$, exactly three $x \in C_{M_{23}}(h)$ satisfy $x^2 = h$ and $x^{-1}rx = r^2$, giving $D_{10} < F_{20}$. The stabilized markings realize the independent (mod 5) multiplier-2 action and the (mod 3) inversion by swapping the unique ordered $2A \cdot 2A \rightarrow 3A$ pair.*

(xii) For a fixed $2A$ element, the ordered factorization spaces satisfy

$$3A \cdot 3A : 1344 + 896 + 224 + 224, \quad 2A \cdot 2A : 84 + 14$$

under the corresponding centralizer actions. The five pair-fusion tails of the branch-cycle-equivariant refinement have arithmetic-genus contributions 4, 4, 2, 4, 4 and explicit rational local models.

(xiii) The zero fiber of β_5 is the finite étale scheme

$$\mathrm{Spec} \mathbb{Q} \sqcup \mathrm{Spec} \mathbb{Q}(\zeta_5).$$

Inversion fixes exactly the rational summand, whose primitive central projector has trace one. On the complementary factor $\mathbb{Q}(\zeta_5)$, inversion has fixed field $\mathbb{Q}(\sqrt{5})$ and defines a quadratic extension $\mathbb{Q}(\zeta_5)/\mathbb{Q}(\sqrt{5})$. After base change to $\mathbb{Q}(\sqrt{5})$, the invariant algebra is the trivalent algebra $N^3 = N$, with the fixed projector e_0 and the two Galois-exchanged projectors $e_{\pm}$.

(xiv) The selected ten-state marking cell $\Delta_{10} = R_{10}$ has q_1 -cycle type 5^2 . Canonical rooting of the two five-cycles selects the fixed-point-free phase pairing κ_{pair} commuting with q_1 and gives

$$\Delta_{10} \cong \mathbb{Z}_5 \times \mathbb{Z}_2, \quad q_1(n, p) = (n + 1, p), \quad \kappa_{\mathrm{pair}}(n, p) = (n, p + 1).$$

The five pairing fibers have cardinality two; $(I \pm \kappa_{\mathrm{pair}})/2$ have rank five. The multiplier-2 lift $X(n, p) = (2n, p)$ has order four, commutes with κ_{pair} , and satisfies $Xq_1X^{-1} = q_1^2$; the resulting action has order 40 and its quotient by $\langle \kappa_{\mathrm{pair}} \rangle$ has order 20 and type F_{20} . The subgroup $\langle q_1, \kappa_{\mathrm{pair}} \rangle$ has order 10 and is transitive, and the algebraic model $\beta_{10}(t) = 1 - (1 - t^2)^5$ quotients through $v = t^2$ to $\beta_5(v)$. The native return $\kappa_{\mathrm{ret}}(x) = d(c(x))$, where $d = q_1q_2q_1q_3q_2$ and c is inverse reversal, preserves Δ_{10} and satisfies

$$\kappa_{\mathrm{ret}}(n, p) = (1 - n, p + 1), \quad \kappa_{\mathrm{ret}}q_1\kappa_{\mathrm{ret}} = q_1^{-1}.$$

Hence, $\langle q_1, \kappa_{\mathrm{ret}} \rangle \cong D_{10}$.

(xv) The mixed inertia q_4^2 has exactly 264 width-11 cycles. Their induced q_1, q_2 action has four transitive components of degree 66. On every component, the branch-cycle triple has cycle types

$$3^{22}, \quad 2^{33}, \quad 34^35^36^6,$$

and genus zero. The involution $\kappa_{11} = (q_1q_2q_3)c$ satisfies $\kappa_{11}q_4^2\kappa_{11} = (q_4^2)^{-1}$, fixes no width-11 cusp, and pairs the four components as $0 \leftrightarrow 2$ and $1 \leftrightarrow 3$.

Proof. Parts (i)–(iv) follow from the finite computations in sections 2, 4 and 5. Part (v) is the coefficient expansion of section 3, part (vi) is the exact class-factorization certificate of theorem 7.3, part (vii) is theorem 7.5, part (viii) is theorem 7.6, and part (ix) is theorems 6.3 and 6.7 to 6.9. Part (x) is theorem 7.8; part (xi) is theorem 7.9; part (xii) is theorems 7.11 and 7.12. Part (xiii) is theorems 7.14 and 7.15; part (xiv) is theorems 6.4 and 6.5; part (xv) is theorem 6.6. The arithmetic divisor statement in (ii) follows from theorem 2.5. $\square$

1.2 Comparison with the three-point realization

	Huang–Jackson–Lee–Poonen– Pries–Zhang	Degree-980 construction
Branch data	$(2, 23A, 23B)$	$(2A, 2A, 3A, 5A), (2A)^4, 3A,$ $(2A, 3A^4),$ and $2A^{10}$ refinements
Branch count	3	4, 5, 6, and 10
Primary finite object	7 Nielsen points	980- and 21,456-state braid components
Primary geometry	degree-23 quotient of genus 4	ordered genus-119 curve and reduced genus-43 quotient
Arithmetic output	explicit regular extension and polynomial	fixed/dual cyclic-fiber arithmetic; exact Hurwitz and collision geometry

The two constructions use different Nielsen data and different finite moduli objects. Huang–Jackson–Lee–Poonen–Pries–Zhang supply an explicit regular realization and polynomial; the degree-980 construction supplies complementary positive-dimensional Hurwitz geometry and a controlled four-to-five boundary correspondence.

2 The four-branch component

Permutations act on the right, so $i^{xy} = (i^x)^y$. Fix the product-one generating tuple

$$\begin{aligned}
g_1 &= (1\ 7)(2\ 20)(4\ 11)(5\ 10)(6\ 18)(9\ 21)(16\ 23)(17\ 22), \\
g_2 &= (1\ 20)(2\ 7)(5\ 10)(6\ 9)(8\ 14)(13\ 19)(17\ 22)(18\ 21), \\
g_3 &= (1\ 21\ 2)(3\ 23\ 14)(4\ 16\ 18)(5\ 8\ 9)(6\ 12\ 22)(13\ 15\ 20), \\
g_4 &= (2\ 6\ 22\ 12\ 21)(3\ 8\ 5\ 18\ 23)(4\ 9\ 14\ 16\ 11)(7\ 20\ 15\ 19\ 13).
\end{aligned}$$

The tuple has class type $(2A, 2A, 3A, 5A)$, generates M_{23} , and has product one.

Theorem 2.1 (Complete ordered Nielsen component). *The ordered inner and absolute Nielsen classes coincide, contain exactly 980 generating classes, and form one pure-braid orbit. The ordered cross-ratio cover is connected of degree 980.*

Computational proof. The primary verifier canonicalizes simultaneous sheet relabeling, closes the printed generating seed under the three pure point-push operators and their inverses, verifies the Artin relations, and obtains one orbit of size 980. Hurwitz moves carry the printed generating tuple to every state in that orbit, so every state generates M_{23} . Independently, the exhaustiveness verifier fixes a $5A$ entry and tests all $3,795^2 = 14,402,025$ ordered pairs from $2A \times 2A$. It finds 14,700 transitive solutions; the centralizer of the fixed $5A$ element has order 15 and acts freely, giving exactly 980 transitive inner classes. Hence, the already constructed 980 generating classes exhaust the transitive Nielsen class. Since $N_{S_{23}}(M_{23}) = M_{23}$ [1], inner and absolute equivalence agree in the natural action. Section 8 lists the scripts and corresponding run logs. $\square$

Corollary 2.2 ($\mathbb{Q}$ -model of the ordered component). *The ordered Hurwitz component of theorem 2.1, its cross-ratio map, and its compactification C are defined over $\mathbb{Q}$.*

Proof. The classes $2A$, $3A$, and $5A$ are rational conjugacy classes of M_{23} [1]. The branch-cycle action of $G_{\mathbb{Q}}$ therefore preserves the ordered Nielsen class. Since theorem 2.1 proves that the complete

generating Nielsen class is one braid orbit, Galois preserves that connected Hurwitz component. Standard Hurwitz-space descent then gives the component, its ordered cross-ratio morphism, and the compactification over $\mathbb{Q}$ [2]. $\square$

Proposition 2.3 (Ordered passport and genus). *The three ordered peripheral permutations have cycle types*

$$1^{24}2^{136}3^{178}5^{30},$$

$$1^22^{21}3^{16}4^{41}5^{46}7^{38}8^{12}11^{12},$$

and the same second type again. Their indices are 612, 792, 792, so the compactified ordered curve C has genus 119.

Proof. The cycle counts are 368, 188, 188. Riemann–Hurwitz gives

$$2g(C) - 2 = -2 \cdot 980 + 612 + 792 + 792 = 236.$$

$\square$

2.1 Boundary divisors and period-index information

For $a \in \{0, 1, \infty\}$ and $e \geq 1$, let $D_{a,e}$ be the sum of geometric points in the compactified fiber over a having ramification index e . By theorem 2.2, each $D_{a,e}$ is $\mathbb{Q}$ -rational because the cross-ratio map and the ramification index are Galois invariant.

Proposition 2.4 (Boundary divisor degrees). *The nonzero degrees are*

a	$e = 1$	$e = 2$	$e = 3$	$e = 4$	$e = 5$	$e = 7$	$e = 8$	$e = 11$
0	24	136	178	0	30	0	0	0
1	2	21	16	41	46	38	12	12
∞	2	21	16	41	46	38	12	12

Theorem 2.5 (Index and period one). *Since $D_{1,1}$ and $D_{1,2}$ are $\mathbb{Q}$ -rational, the divisor*

$$\Delta = 11D_{1,1} - D_{1,2}$$

is $\mathbb{Q}$ -rational and has degree one. Consequently,

$$\text{ind}_{\mathbb{Q}}(C) = \text{per}_{\mathbb{Q}}(C) = 1, \quad \text{Pic}^1(C)(\mathbb{Q}) \neq \emptyset.$$

Proof. The degrees in theorem 2.4 give $\deg \Delta = 11 \cdot 2 - 21 = 1$. The degrees of rational zero-cycles generate the index subgroup, while the degree of rational divisor classes generates the period subgroup. Both subgroups therefore contain 1. $\square$

Theorem 2.6 (Rational boundary points). *The compactified ordered curve has eight individually $\mathbb{Q}$ -rational boundary points. In the one-based canonical enumeration, they are*

$$281, 372, 823, 837 \quad (\lambda = 1), \quad 20, 770, 818, 973 \quad (\lambda = \infty).$$

The width-one divisors $D_{1,1}$ and $D_{\infty,1}$ are spectra of imaginary quadratic fields. Consequently, $C(\mathbb{Q}) \neq \emptyset$.

Computational proof. The exact real-boundary verifier constructs the unique complex-conjugation permutation on the ordered Nielsen fiber, with cycle type $1^{26}2^{477}$, and computes boundary fixed-point counts 26, 28, 28. The stable boundary signature records ramification index, node monodromy, component monodromy groups, and degree-23 orbit partitions. Exactly eight real boundary points have signatures occurring once in their rational boundary fiber, so Galois invariance of the signature fixes each of those eight points individually. The degree-two width-one divisors over $\lambda = 1$ and $\lambda = \infty$ have empty real support, so each is an imaginary quadratic field. $\square$

Remark 2.7 (Boundary and interior arithmetic). Index one records the rational divisor-class statement. The eight singleton boundary signatures separately certify $C(\mathbb{Q}) \neq \emptyset$. Interior specialization is governed by the fiber-algebra criterion in theorem 3.3.

3 Universal coefficient reconstruction targets

3.1 Four branches

Normalize the branch values by $3A \mapsto 0$, $2A \mapsto 1$, $5A \mapsto \infty$, and the second $2A \mapsto \lambda$. Put

$$\begin{aligned} A_3 &= x^6 + a_4x^4 + a_3x^3 + a_2x^2 + a_1x + a_0, & A_1 &= x^5 + \alpha_4x^4 + \cdots + \alpha_0, \\ Q_5 &= x^4 + q_2x^2 + q_1x + q_0, & Q_1 &= r_3x^3 + r_2x^2 + r_1x + r_0, \\ B_2 &= x^8 + b_6x^6 + \cdots + b_0, & B_1 &= s_7x^7 + \cdots + s_0, \\ C_2 &= x^8 + c_7x^7 + \cdots + c_0, & C_1 &= t_7x^7 + \cdots + t_0. \end{aligned}$$

Set $p = A_3^3A_1$ and $q = Q_5^5Q_1$. The identities

$$p - q = B_2^2B_1, \quad p - \lambda q = C_2^2C_1$$

give 48 coefficient equations in 48 normalized coefficient variables.

Proposition 3.1 (Meaning of the four-branch equations). *On the locus where all factors are square-free, pairwise disjoint, and of the displayed degrees, every solution defines a degree-23 rational map with local cycle partitions 3^61^5 , 2^81^7 , 5^41^3 , and 2^81^7 at $(0, 1, \infty, \lambda)$. Conversely, every normalized M_{23} -cover in the selected Hurwitz component gives a solution. The component of the coefficient locus containing the certified Nielsen tuple has geometric monodromy M_{23} ; the raw factorization equations encode the full ramification locus, and the Nielsen data select the M_{23} component.*

Proof. The exponents 3, 2, 5, 2 encode the ramification multiplicities in the four fibers; the residual factors encode the unramified points. Branch separation and nonvanishing resultants make the rational map nondegenerate. Reading the fiber factorizations in reverse gives the converse. $\square$

Remark 3.2. The certificate bundle expands all 48 equations exactly. The equations encode the normalized ramification reconstruction problem, and the Nielsen computation supplies the complete component selection used in the geometric results below.

3.2 Interior fiber algebra and the degree-27 target

Let $a \in \mathbb{P}_\lambda^1 \setminus \{0, 1, \infty\}$ be rational and let A_a be the finite étale $\mathbb{Q}$ -algebra of the degree-980 ordered fiber above a .

Theorem 3.3 (Interior fiber criterion and target normal form). *For a rational interior parameter a , the degree-980 fiber contains a $\mathbb{Q}$ -rational point exactly when its finite étale algebra A_a has a*

direct $\mathbb{Q}$ -factor, equivalently a primitive central idempotent of multiplication rank and trace one. The executed target algebra

$$B_{27} = \mathbb{Q}[Z]/(Z^{27} - Z) \simeq \mathbb{Q}^3 \times \mathbb{Q}(\zeta_{13})^2$$

has rank 27, characteristic polynomial $T^{27} - T$, and three rational rank-one trace-one projectors; the zero-factor projector is $e_0 = 1 - Z^{26}$. A certified map $B_{27} \rightarrow A_a$ carrying e_0 to such a projector, or a certified surjection $A_a \twoheadrightarrow B_{27}$ followed by $Z \mapsto 0$, supplies an interior rational point. The release records B_{27} and its exact projectors as the target normal form; source activation consists of the genuine algebra A_a and the certified algebra map.

Proof. Finite étale $\mathbb{Q}$ -algebras split as products of number fields. The rational factor and trace-one projector criteria are therefore equivalent. The Chinese remainder factorization of $Z^{27} - Z$ gives the displayed decomposition, and $1 - Z^{26}$ is the projector to the root $Z = 0$. $\square$

3.3 Five branches

Normalize the five branch values to $(0, \infty, 1, \lambda, \mu)$. Write

$$\begin{aligned} P &= A_0^2 L_0, & Q &= A_\infty^2 L_\infty, \\ P - Q &= A_1^2 L_1, & P - \lambda Q &= A_\lambda^2 L_\lambda, & P - \mu Q &= C^3 M, \end{aligned}$$

with $\deg A_\bullet = 8$, $\deg L_\bullet = 7$, $\deg C = 6$, and $\deg M = 5$. The three degree-23 identities give 72 coefficient equations. Before gauge reduction there are 83 variables; the five factor rescalings, common target rescaling, and three source-coordinate dimensions account for nine gauge dimensions, leaving expected base dimension two.

Proposition 3.4 (Meaning of the five-branch equations). *On the nondegenerate locus, every solution defines a degree-23 cover of source genus zero with four local cycle partitions $2^8 1^7$ and one partition $3^6 1^5$. Conversely, every normalized M_{23} -cover in the selected five-branch component supplies a solution. As in the four-branch case, the raw equations encode ramification and the Nielsen data select the M_{23} component.*

Proof. The four square factors and one cube factor encode indices 8, 8, 8, 8, 12, whose sum is $44 = 2(23 - 1)$. The residual factors encode the unramified portions. The same factorization argument as in theorem 3.1 proves both directions. $\square$

4 The reduced j -line

The 980-sheet ordered cross-ratio cover records one class ordering. Passing through all twelve orderings and quotienting by the certified free Klein-four action produces the reduced j -line cover.

Theorem 4.1 (Reduced j -line certificate). *The full all-orderings orbit has 11,760 states. The Klein-four quotient has degree 2940 and one connected component. Its branch-cycle triple has cycle types*

$$\begin{aligned} \gamma_0 &\sim 3^{980}, & \gamma_1 &\sim 2^{1470}, \\ \gamma_\infty &\sim 2^{14} 3^{38} 4^{89} 5^{30} 6^{86} 8^{41} 10^{46} 14^{38} 16^{12} 22^{12}. \end{aligned}$$

The compactification has genus 43 and 406 cusps.

Computational proof. The all-orderings verifier checks the braid relations, twelve orderings, free Klein-four action, transitivity, and the three quotient permutations. Their indices are 1960, 1470, 2534. Thus,

$$2g - 2 = -2 \cdot 2940 + 1960 + 1470 + 2534 = 84.$$

The number of cycles of γ_∞ is 406. □

Corollary 4.2 (Elliptic fiber over $j = 1728$). *Every point over $j = 1728$ has ramification index two; equivalently, the reduced elliptic fiber is fixed-point-free of type 2^{1470} .*

5 The five-branch component and the width-five boundary

An explicit product-one generating seed of type $(2A, 2A, 2A, 2A, 3A)$ lies in the standard M_{23} copy. The colored braid action of q_1, q_2, q_3, q_4^2 closes its component exactly.

Theorem 5.1 (Five-branch colored component). *The colored braid component has 21,456 states. Each of q_1, q_2, q_3 has cycle type*

$$1^{24} 2^{292} 3^{796} 4^{1248} 5^{980} 6^{1428},$$

while q_4^2 has cycle type

$$1^{18} 2^{468} 3^{582} 4^{984} 5^{900} 7^{840} 8^{192} 11^{264}.$$

Computational proof. The verifier reconstructs the component from the explicit seed, checks product one, source genus zero, generation of M_{23} , operator bijectivity, and every displayed cycle histogram. □

Theorem 5.2 (Full equal-2A width-five boundary). *Let $D_{ij} \subset \widetilde{\mathcal{H}}_5$, $1 \leq i < j \leq 4$, be the six collision divisors for the four equal 2A branch points. Every width-five stratum $D_{ij}^{(5)}$ has 980 cycles, clutching maps these cycles bijectively onto $\text{Ni}^{\text{in}}(M_{23}; 2A, 2A, 3A, 5A)$, and the normalized stratum is the complete ordered genus-119, degree-980 four-branch curve $C^{\text{ord}}/\mathbb{Q}$. Hence, $\mathcal{B}_{\text{full}} = \bigcup_{i < j} D_{ij}^{(5)}$ is the full equal-2A width-five boundary subcover.*

Computational and geometric proof. For D_{12}, D_{23}, D_{34} the local operators are q_1, q_2, q_3 . The verifier gives 980 width-five cycles and 980 distinct clutched transitive inner classes for each operator. By theorem 2.1, the complete four-branch Nielsen class has 980 elements, so these three clutching maps are bijections.

The four labels carry the same rational class $2A$, and the colored braid action induces the full S_4 permutation action on them while preserving the connected component of theorem 5.1. Every pair $\{i, j\}$ is therefore the image of $\{1, 2\}$ under a colored braid. Local monodromy is transported by conjugation and clutching is braid-equivariant, giving the same 980-cycle bijection for all six pairs. Admissible-cover clutching gives a finite degree-one morphism over $\mathbb{Q}$ from each normalized width-five stratum to C^{ord} [10]; normality then gives the stated isomorphism. □

6 The degree-five cyclic quotient and fixed/dual fiber

The cyclic degree-five carrier associated with the five-pair fusion has abstract branch cycles

$$\sigma_0 = 1, \quad \sigma_1 = (1\ 2\ 3\ 4\ 5), \quad \sigma_\infty = \sigma_1^{-1}.$$

Proposition 6.1 (Abstract five-pair cover). *The branch cycles above are transitive, have product one, and have total index eight. Their normalized cover is*

$$\bar{L} \cong \mathbb{P}_{\mathbb{Q}}^1, \quad \boxed{\beta_5(t) = 1 - (1 - t)^5}.$$

The fiber $\beta_5^{-1}(0)$ is a $\mathbb{Q}$ -rational effective zero-cycle of degree five.

Proof. Riemann–Hurwitz gives $2g - 2 = -10 + 8 = -2$. The displayed polynomial has one totally ramified point over 1, one over ∞ , and five unramified points over 0. $\square$

Proposition 6.2 (Finite cyclic carrier of the pair fusion). *Let $P = (P_1, \dots, P_5)$ be the fused five-branch tuple of theorem 7.8 and set*

$$\delta = q_1 q_2 q_3 q_4.$$

Modulo simultaneous conjugation, the five classes

$$[P], [\delta P], [\delta^2 P], [\delta^3 P], [\delta^4 P]$$

are distinct and $[\delta^5 P] = [P]$. Hence, δ acts as a 5-cycle on this finite carrier. The associated abstract degree-five branch-cycle triple has passport $1^5, 5, 5$ and therefore the normalized model β_5 of theorem 6.1. This identifies the β_5 passport with the pair-fusion carrier at the Nielsen level. The corresponding global admissible-cover geometry is the pair-marked clutching construction of theorem 7.18.

Exact computational proof. The certificate reconstructs P from the branch-cycle-equivariant ten-involution refinement, applies $q_1 q_2 q_3 q_4$ successively, canonicalizes only by simultaneous conjugation, and obtains five distinct inner keys before the initial key returns on the fifth iterate. The resulting permutation is a 5-cycle, so the branch-cycle triple $1, \delta, \delta^{-1}$ has passport $1^5, 5, 5$. $\square$

Proposition 6.3 (Mixed-boundary monodromy). *Let D_{45} denote the mixed collision boundary whose normal inertia on the 21,456-state colored component is q_4^2 . Its fixed fiber*

$$F_{45} = \text{Fix}(q_4^2)$$

has 18 states and is one orbit under the boundary monodromy generated by q_1, q_2 . On F_{45} , the cycle profiles are

$$\text{ct}(q_1) = 2^1 3^2 5^2, \quad \text{ct}(q_2) = 2^1 3^2 5^2, \quad \text{ct}((q_1 q_2)^{-1}) = 3^6.$$

Hence, the normalized boundary cover has degree 18 and genus 2. The centralizer of this transitive boundary monodromy has order 2 and its nontrivial element is fixed-point-free.

Let $R_{10} \subset F_{45}$ be the ten-state subset consisting of the q_4^2 -fixed sheets lying on the width-five cycles of q_1 . Then

$$|R_{10}| = 10, \quad q_2(R_{10}) \neq R_{10}.$$

The boundary monodromy therefore moves R_{10} through a finite orbit of ten-element subsets. The canonical set-stabilizer base change associated with this orbit produces the connected degree-10 Hurwitz subcover of theorem 6.7. The exact cover β_5 of theorem 6.1 records the cyclic five-pair quotient on the distinguished fiber.

Exact computational proof. The boundary verifier reconstructs the complete five-branch orbit, restricts q_1, q_2 to F_{45} , and proves transitivity on all 18 fixed sheets. Their cycle counts are 5, 5, 6, so the three indices are 13, 13, 12 and

$$g = 1 - 18 + \frac{13 + 13 + 12}{2} = 2.$$

Direct centralizer enumeration gives order 2 and a unique free involution. The same verifier constructs R_{10} and its q_2 -image and checks that the two ten-element sets differ. The subset-orbit and stabilizer computation in theorem 6.7 then supplies the canonical finite base change on which R_{10} becomes a connected Hurwitz subcover. $\square$

Theorem 6.4 (Dual two-five marking ledger). *Let*

$$\Delta_{10} := R_{10}.$$

The restriction of q_1 to Δ_{10} has cycle type 5^2 , while q_4^2 acts trivially. Root each q_1 -cycle at its lexicographically least canonical Nielsen representative and write the two rooted cycles as

$$\Delta_5^0 = (P_{0,0}, \dots, P_{4,0}), \quad \Delta_5^1 = (P_{0,1}, \dots, P_{4,1}),$$

with $q_1 P_{n,p} = P_{n+1,p}$, indices modulo 5. Among the five phase-compatible free involutions exchanging the two cycles and commuting with q_1 , this rooting selects the unique involution

$$\boxed{\kappa_{\text{pair}}(P_{n,p}) = P_{n,p+1}.$$

Consequently,

$$\boxed{\Delta_{10} \simeq \mathbb{Z}_5 \times \mathbb{Z}_2, \quad q_1(n, p) = (n+1, p), \quad \kappa_{\text{pair}}(n, p) = (n, p+1),}$$

and the dual-forgotten projection $(n, p) \mapsto n$ has fiber profile

$$\boxed{2, 2, 2, 2, 2.}$$

On $\mathbb{Q}^{\Delta_{10}}$, the operators

$$P_+ = \frac{I + \kappa_{\text{pair}}}{2}, \quad P_- = \frac{I - \kappa_{\text{pair}}}{2}$$

are orthogonal idempotents of ranks 5 and 5. The ledger multiplier

$$X(n, p) = (2n, p)$$

has order four, commutes with κ_{pair} , and satisfies

$$X q_1 X^{-1} = q_1^2.$$

Thus, $\langle q_1, \kappa_{\text{pair}}, X \rangle$ has order 40, and its quotient by the central dual involution has order 20 and is isomorphic to $F_{20} = C_5 \rtimes C_4$.

The same two-five ledger has the genus-zero algebraic model

$$\boxed{\beta_{10}(t) = 1 - (1 - t^2)^5, \quad \kappa_{\text{pair}}(t) = -t, \quad v = t^2, \quad \beta_{10}(t) = \beta_5(v).}$$

For the generic fiber polynomial $\beta_{10}(t) - y$,

$$\text{Disc}_t(\beta_{10}(t) - y) = 10^{10} y (y - 1)^8,$$

so the rank-ten algebra is finite étale over $\mathbb{Q}[y, 1/(y(y-1))]$, and the κ_{pair} -invariant algebra has rank five.

Exact computational proof. The exact colored-orbit reconstruction produces the 21,456 states and the selected cell

$$\{5619, 5707, 10252, 11406, 13499, 15797, 17032, 18695, 20064, 21190\}$$

in zero-based enumeration. Restriction of q_1 gives two five-cycles. Canonical rooting gives

$$\begin{aligned} \Delta_5^0 &= (21190, 11406, 17032, 15797, 20064), \\ \Delta_5^1 &= (5707, 10252, 13499, 18695, 5619), \end{aligned}$$

and hence the five canonical dual pairs

$$(21190, 5707), (11406, 10252), (17032, 13499), (15797, 18695), (20064, 5619).$$

Direct permutation arithmetic verifies $\kappa_{\text{pair}}^2 = 1$, fixed-point-freeness, $q_1 \kappa_{\text{pair}} = \kappa_{\text{pair}} q_1$, the 2^5 projected-key fibers, the two rank-five projectors, the multiplier identity $X q_1 X^{-1} = q_1^2$, and the stated group orders. Symbolic elimination gives the displayed discriminant and the quotient identity $\beta_{10}(t) = \beta_5(t^2)$. $\square$

Theorem 6.5 (Native return on the ten-state cell). *Let c be inverse reversal on the five-branch component and, in the recorded right-action convention, set*

$$d = q_1 q_2 q_1 q_3 q_2, \quad \boxed{\kappa_{\text{ret}}(x) = d(c(x))}.$$

Then κ_{ret} preserves Δ_{10} , is a fixed-point-free involution, and in the rooted coordinates of theorem 6.4 acts by

$$\boxed{\kappa_{\text{ret}}(n, p) = (1 - n, p + 1)}.$$

Moreover,

$$\boxed{\kappa_{\text{ret}} q_1 \kappa_{\text{ret}} = q_1^{-1}},$$

so

$$\boxed{\langle q_1, \kappa_{\text{ret}} \rangle \cong D_{10}}.$$

The phase pairing and native return commute,

$$\kappa_{\text{pair}} \kappa_{\text{ret}} = \kappa_{\text{ret}} \kappa_{\text{pair}},$$

and $\langle q_1, \kappa_{\text{pair}}, \kappa_{\text{ret}} \rangle$ has order 20.

Exact computational proof. The completion certificate reconstructs the same 21,456-state component and the ten-state cell Δ_{10} . Applying inverse reversal and then the braid word d gives the five pairs

$$(5619, 17032), (5707, 11406), (10252, 21190), (13499, 20064), (15797, 18695).$$

These pairs preserve Δ_{10} and contain no fixed state. Relative to the canonical rooted cycles

$$\begin{aligned} \Delta_5^0 &= (21190, 11406, 17032, 15797, 20064), \\ \Delta_5^1 &= (5707, 10252, 13499, 18695, 5619), \end{aligned}$$

the five pairs are exactly $(n, p) \leftrightarrow (1 - n, p + 1)$. Direct permutation arithmetic gives the conjugation identity with q_1 , commutation with κ_{pair} , and the group orders 10 and 20. $\square$

Theorem 6.6 (Width-eleven mixed-boundary components). *Let $\mathcal{C}_{11}$ be the set of length-11 cycles of the mixed inertia q_4^2 on the 21,456-state five-branch component. Then*

$$\boxed{|\mathcal{C}_{11}| = 264}.$$

The induced action of q_1 and q_2 splits $\mathcal{C}_{11}$ into four transitive components

$$\mathcal{B}_0, \mathcal{B}_1, \mathcal{B}_2, \mathcal{B}_3, \quad |\mathcal{B}_i| = 66.$$

On each component, define

$$\gamma_0 = q_1 q_2, \quad \gamma_1 = q_1 q_2 q_1, \quad \gamma_\infty = (\gamma_1 \gamma_0)^{-1}$$

in the recorded right-action convention. Their cycle types are

$$\boxed{\text{ct}(\gamma_0) = 3^{22}, \quad \text{ct}(\gamma_1) = 2^{33}, \quad \text{ct}(\gamma_\infty) = 3 \, 4^3 5^3 6^6.}$$

Every normalized degree-66 component has genus zero.

Let c be inverse reversal and define

$$\boxed{\kappa_{11}(x) = (q_1 q_2 q_3)(c(x)).}$$

Then κ_{11} is an involution satisfying

$$\boxed{\kappa_{11} q_4^2 \kappa_{11} = (q_4^2)^{-1}.}$$

It fixes no width-11 cusp and pairs the four components as

$$\boxed{0 \leftrightarrow 2, \quad 1 \leftrightarrow 3.}$$

Exact computational proof. The completion certificate finds exactly 264 length-11 cycles of q_4^2 . The induced q_1, q_2 action has orbit sizes 66, 66, 66, 66. On every orbit the computed cycle histograms are 3^{22} , 2^{33} , and $3 \, 4^3 5^3 6^6$, with indices 44, 33, and 53. Riemann–Hurwitz gives

$$2g - 2 = -132 + 44 + 33 + 53 = -2, \quad g = 0.$$

The same certificate constructs κ_{11} , verifies $\kappa_{11}^2 = 1$ and the inverse-inertia identity, counts zero fixed width-11 cusps, and obtains the stated component pairing. $\square$

Theorem 6.7 (Canonical ten-sheet pullback over the mixed boundary). *Let $B^\circ \simeq M_{0,4,\mathbb{Q}}$ be the smooth base of the mixed boundary divisor D_{45} and let*

$$\rho : \pi_1(B_{\mathbb{Q}}^\circ) \longrightarrow S(F_{45})$$

be the geometric monodromy representation of the degree-18 boundary cover. Write

$$G = \langle a, b \rangle, \quad a = q_1|_{F_{45}}, \quad b = q_2|_{F_{45}},$$

and let $\Delta_{10} = R_{10}$. Then

$$|G| = 92,897,280, \quad Z(G) = \langle \kappa_{18} \rangle \cong C_2,$$

where κ_{18} is the unique nontrivial deck transformation of the degree-18 boundary cover and acts freely on F_{45} .

The G -orbit of Δ_{10} in the set of ten-element subsets of F_{45} has cardinality 126. Hence,

$$H := \text{Stab}_G(\Delta_{10})$$

has order 737,280. The induced action of H on Δ_{10} is transitive of order 3840. The involution κ_{18} preserves Δ_{10} and its restriction is exactly the involution κ_{pair} of theorem 6.4. On the five κ_{pair} -pairs, the induced H -action has order 120 and equals S_5 .

Let

$$B_H^\circ \longrightarrow B^\circ$$

be the connected finite étale cover corresponding to $\rho^{-1}(H)$. The pullback of the degree-18 boundary Hurwitz cover to B_H° contains a connected degree-10 subcover

$$L_{10,H}^\circ \longrightarrow B_H^\circ$$

with fiber Δ_{10} . The involution κ_{pair} is a deck involution of this subcover, and

$$Q_H^\circ := L_{10,H}^\circ / \langle \kappa_{\text{pair}} \rangle \longrightarrow B_H^\circ$$

is a connected degree-5 cover.

After compactification, the three curves have

$$\boxed{g(\overline{B}_H) = 31, \quad g(\overline{L}_{10,H}) = 327, \quad g(\overline{Q}_H) = 161.}$$

As covers of the original compactified mixed-boundary base $\mathbb{P}^1$, their degrees are 126, 1260, and 630. The involution κ_{pair} acts freely on $L_{10,H}^\circ$; the compactified quotient map $p_H : \overline{L}_{10,H} \rightarrow \overline{Q}_H$ has total boundary ramification 12.

Exact finite-monodromy and covering-space proof. The mixed-boundary certificate gives the restricted permutations a, b on 18 sheets. Schreier–Sims computation gives $|\langle a, b \rangle| = 92,897,280$. Its permutation centralizer has order two; the nonidentity element lies in the monodromy group, commutes with a and b , and acts freely. The pullback certificate enumerates the orbit of Δ_{10} under G and obtains 126 subsets. Orbit–stabilizer gives $|H| = 737,280$. Schreier generators for H induce a transitive group of order 3840 on Δ_{10} and a transitive group of order 120 on the five κ_{pair} -pairs. The restriction of the central deck involution gives the five pairs

$$(5619, 20064), (5707, 21190), (10252, 11406), (13499, 17032), (15797, 18695),$$

which agree exactly with the canonical pairing of theorem 6.4.

Finite étale covers of $B_{\mathbb{Q}}^\circ$ correspond to finite-index subgroups of $\pi_1(B_{\mathbb{Q}}^\circ)$. The cover associated with $\rho^{-1}(H)$ makes Δ_{10} monodromy-invariant, and transitivity of the induced H -action makes the resulting degree-10 component connected. Centrality of κ_{pair} gives the degree-5 quotient.

For the compactified degree-126 base-change cover, the three branch permutations have cycle types

$$1^1 5^4 15^7, \quad 1^1 5^4 15^7, \quad 3^{42}.$$

Their indices are 114, 114, 84, giving $g(\overline{B}_H) = 31$. On the degree-1260 ten-sheet component, the corresponding types are

$$5^{20} 10^2 15^{68} 30^4, \quad 5^{20} 10^2 15^{68} 30^4, \quad 3^{420},$$

with indices 1166, 1166, 840, giving genus 327. On the degree-630 quotient, they are

$$5^{12} 15^{38}, \quad 5^{12} 15^{38}, \quad 3^{210},$$

with indices 580, 580, 420, giving genus 161. Finally, Riemann–Hurwitz for p_H gives

$$(2 \cdot 327 - 2) - 2(2 \cdot 161 - 2) = 12,$$

so all ramification of the involution quotient lies on the compactification boundary and has total degree 12. $\square$

Lemma 6.8 (Arithmetic stability of the ten-state pullback). *The orbit of Δ_{10} , the stabilizer cover $B_H^\circ \rightarrow B^\circ$, the connected degree-10 component $L_{10,H}^\circ$, the involution κ_{pair} , and the quotient Q_H° are defined over $\mathbb{Q}$. Their normal compactifications and induced finite maps are defined over $\mathbb{Q}$ as well.*

Proof. For the rational tangential basepoint,

$$\Delta_{10} = \{x \in F_{45} : q_4^2 x = x \text{ and } |\langle q_1 \rangle x| = 5\}.$$

The certificate gives exactly ten elements. By the branch-cycle action, q_1 and q_4^2 are carried to simultaneous conjugates of unit powers, which preserve the defining predicate up to the same fiber relabeling [2]. Thus, every Galois conjugate of Δ_{10} is a G -translate of Δ_{10} , and the certified 126-element orbit $G \cdot \Delta_{10}$ is $G_{\mathbb{Q}}$ -stable. The finite etale cover with this orbit as fiber, together with its tautological degree-10 subcover, therefore descends to $\mathbb{Q}$. The unique nontrivial central deck involution is also $G_{\mathbb{Q}}$ -stable; hence, its restriction, the degree-5 quotient, and the induced maps descend. Normalization and the canonical proper compactifications preserve the descent [9]. $\square$

Theorem 6.9 (Comparison morphism). *Let $\overline{D}_{45}$ be the normalized compactified mixed-boundary Hurwitz curve, let*

$$\overline{R}_H : \overline{L}_{10,H} \longrightarrow \overline{D}_{45}$$

be the finite morphism induced by the second projection of the canonical set-stabilizer pullback, and let

$$p_H : \overline{L}_{10,H} \longrightarrow \overline{Q}_H := \overline{L}_{10,H} / \langle \kappa_{\text{pair}} \rangle$$

be the quotient by the deck involution κ_{pair} , which acts fixed-point-freely on $L_{10,H}^\circ$ and has total boundary ramification 12 on the compactification. The unique nontrivial central deck involution κ_{18} of $\overline{D}_{45}$ satisfies

$$\boxed{\overline{R}_H \circ \kappa_{\text{pair}} = \kappa_{18} \circ \overline{R}_H.}$$

Consequently, there is a unique finite $\mathbb{Q}$ -morphism

$$\boxed{\mathfrak{B}_H : \overline{Q}_H \longrightarrow \overline{D}_{45} / \langle \kappa_{18} \rangle}$$

such that, for the quotient map

$$\pi_{45} : \overline{D}_{45} \longrightarrow \overline{D}_{45} / \langle \kappa_{18} \rangle,$$

one has

$$\boxed{\mathfrak{B}_H \circ p_H = \pi_{45} \circ \overline{R}_H.}$$

Equivalently, on geometric points,

$$\boxed{\mathfrak{B}_H([x]_{\kappa_{\text{pair}}}) = [\overline{R}_H(x)]_{\kappa_{18}}.}$$

The diagram

$$\begin{array}{ccc} \overline{L}_{10,H} & \xrightarrow{\overline{R}_H} & \overline{D}_{45} \\ p_H \downarrow & & \downarrow \pi_{45} \\ \overline{Q}_H & \xrightarrow{\mathfrak{B}_H} & \overline{D}_{45} / \langle \kappa_{18} \rangle \end{array}$$

commutes. On the smooth locus, $\mathfrak{B}_H$ records the paired ten-state Hurwitz family after the canonical degree-126 base change.

Proof. By theorem 6.8, the source, target, involutions, quotient maps, and compactifications in the statement are defined over $\mathbb{Q}$. The degree-10 component $L_{10,H}^\circ$ is a connected component of the pullback of the degree-18 mixed-boundary Hurwitz cover. Projection to that cover gives R_H° , and normalization followed by compactification gives $\overline{R}_H$. The pullback certificate verifies that the central deck involution κ_{18} preserves the selected ten-state fiber and restricts there exactly to the canonical pairing κ_{pair} . Centrality makes this identity compatible with monodromy on the connected pullback component, hence

$$\overline{R}_H \kappa_{\text{pair}} = \kappa_{18} \overline{R}_H.$$

Applying π_{45} gives

$$\pi_{45} \overline{R}_H \kappa_{\text{pair}} = \pi_{45} \kappa_{18} \overline{R}_H = \pi_{45} \overline{R}_H.$$

Thus, $\pi_{45}\overline{R}_H$ is constant on the κ_{pair} -orbits. The universal property of the finite quotient p_H , applied over $\mathbb{Q}$, gives the unique $\mathbb{Q}$ -morphism $\mathfrak{B}_H$ and the displayed identity. $\square$

Corollary 6.10 (Closure of the comparison step). *The canonical five-pair quotient is an algebraic curve equipped with the comparison morphism*

$$\boxed{\overline{Q}_H \xrightarrow{\mathfrak{B}_H} \overline{D}_{45}/\langle \kappa_{18} \rangle.}$$

Its compactified genus is 161, and the induced action on the five κ_{pair} -pairs is S_5 . On the distinguished ten-state fiber, the subgroup $\langle q_1, \kappa_{\text{pair}} \rangle$ has order 10; quotienting that cyclic subcarrier by κ_{pair} gives the degree-5 passport

$$1^5, \quad 5, \quad 5,$$

represented by

$$\boxed{\beta_5(t) = 1 - (1 - t)^5.}$$

Thus, the global comparison curve $\overline{Q}_H$ and the genus-zero cyclic carrier $\overline{L} \cong \mathbb{P}_{\mathbb{Q}}^1$ realize two levels of the same five-pair data: $\overline{Q}_H$ carries the full global S_5 monodromy after the canonical base change, while $\overline{L}$ records the cyclic order-5 subcarrier selected by $\langle q_1, \kappa_{\text{pair}} \rangle$.

Proposition 6.11 (Geometric deck pairing and representative duality). *The involution κ_{pair} in theorem 6.7 pairs distinct five-branch Hurwitz sheets. The h -equivariant duality of theorem 7.8 sends the printed representatives R, U, P to simultaneous h -conjugates and therefore fixes their inner Nielsen classes. Thus, the two involutions act on different objects: κ_{pair} is a deck transformation of a Hurwitz cover after the canonical base change, while h -duality is a representative symmetry inside one inner Nielsen point.*

Proof. The five κ_{pair} -pairs listed above consist of ten distinct canonical state indices. The dual-conjugacy certificate verifies

$$R^\vee = hRh^{-1}, \quad U^\vee = hUh^{-1}, \quad P^\vee = hPh^{-1}$$

and equality of the corresponding canonical inner keys. These two exact computations establish the stated type separation. $\square$

Corollary 6.12 (Cyclic carrier inside the canonical pullback). *On Δ_{10} the subgroup $\langle q_1, \kappa_{\text{pair}} \rangle$ has order 10 and acts transitively. Its quotient by $\langle \kappa_{\text{pair}} \rangle$ is cyclic of order five and gives the finite passport $1^5, 5, 5$ represented by β_5 . The canonical global quotient $\overline{Q}_H$ carries the full induced S_5 action and has genus 161, while $\overline{L} \cong \mathbb{P}_{\mathbb{Q}}^1$ is the genus-zero cyclic carrier of theorem 6.1. The two constructions therefore provide complementary global and cyclic realizations of the same five-pair finite data.*

Corollary 6.13 (Infinite tower of finite carriers). *The iterates $\beta_{10}^{\circ n} : \mathbb{P}^1 \rightarrow \mathbb{P}^1$ have degree 10^n for every $n \geq 1$ and therefore form an explicit infinite tower of finite covers. At the first carrier level, $t \mapsto -t$ gives the degree-two dual quotient to β_5 .*

Proof. Degrees multiply under composition, and $\deg \beta_{10} = 10$. $\square$

Remark 6.14 (Finite dual-two-five carrier). The ten-state cell carries a cyclic coordinate together with two explicitly certified involutions. The phase pairing κ_{pair} exchanges equal phases across the two q_1 -cycles and underlies the canonical set-stabilizer pullback. The native return κ_{ret} reverses the q_1 orientation and gives the dihedral action of theorem 6.5. The full 107,280-state simultaneous-collision computation has five orbits of size 21,456, while theorem 6.6 supplies the global genus-zero width-11 boundary decomposition.

Corollary 6.15 (Mixed-boundary and cyclic realizations). *The degree-18 mixed-boundary cover, the canonical degree-10 pullback of theorem 6.7, the genus-zero cyclic carrier of theorem 6.1, and the four width-11 components of theorem 6.6 give complementary global and finite realizations of the same five-branch geometry. The phase-pairing quotient records the cyclic order-five carrier, the native return records the dihedral orientation reversal, and the width-11 components record the global mixed-collision geometry.*

7 Common refinements from three through ten branches

The explicit three-point realization and the four-/five-branch components admit two exact refinement diagrams:

$$\begin{aligned}\mathcal{H}'_5(2A, 3A, 3A, 3A, 3A) &\longrightarrow \mathcal{H}_3(2A, 23A, 23B), \\ \mathcal{H}_5(2A, 2A, 2A, 2A, 3A) &\longrightarrow \mathcal{H}_4(2A, 2A, 3A, 5A).\end{aligned}$$

The second arrow is the width-five clutching map of theorem 5.2. This section constructs the first arrow exactly, joins the two five-branch components through the class-restricted minimal six-branch refinement, upgrades both collision faces to explicit algebraic tail models over $\mathbb{Q}$, and then constructs the separate ten-involution refinement layer. To preserve the released permutation tuple verbatim, theorem 7.3 uses the left-action convention of [4]; the accompanying certificate states and verifies that convention explicitly.

Proposition 7.1 (Codimension-one four-to-three collision obstruction). *A single collision of the class vector $(2A, 2A, 3A, 5A)$ produces at most one conjugacy class absent from the original vector, whereas the target triple $(2A, 23A, 23B)$ requires both $23A$ and $23B$.*

Proof. A single collision replaces two adjacent entries by one product and leaves two entries unchanged. It can therefore introduce at most one conjugacy class absent from the original vector. The target triple requires the two new classes $23A$ and $23B$, establishing the obstruction. $\square$

Lemma 7.2 (Two-involution product obstruction for 23-elements). *Products of two involutions in M_{23} avoid the classes $23A$ and $23B$.*

Proof. Inversion exchanges the classes $23A$ and $23B$ [1, 4]. If $g = ab$ with $a^2 = b^2 = 1$, then

$$aga = a(ab)a = ba = (ab)^{-1} = g^{-1}.$$

Thus, g would be conjugate to g^{-1} in M_{23} , contrary to the two distinct inverse 23-classes. $\square$

Theorem 7.3 (Exact five-branch refinement of the released triple). *Let (g_1, g_2, g_3) be the explicit product-one tuple of [4], with class vector $(2A, 23A, 23B)$. There are 3A elements a_2, b_2, a_3, b_3 such that*

$$g_2 = a_2 b_2, \quad g_3 = a_3 b_3,$$

and the five-tuple

$$(g_1, a_2, b_2, a_3, b_3)$$

has class vector $(2A, 3A, 3A, 3A, 3A)$, product one, and generated group M_{23} . One exact choice is

$$\begin{aligned}a_2 &= (1\ 2\ 15)(3\ 12\ 7)(4\ 8\ 18)(6\ 23\ 14)(9\ 10\ 16)(11\ 21\ 17), \\ b_2 &= (2\ 17\ 18)(3\ 6\ 7)(5\ 8\ 22)(9\ 14\ 11)(12\ 15\ 13)(19\ 20\ 23), \\ a_3 &= (1\ 9\ 16)(2\ 18\ 8)(3\ 7\ 22)(4\ 10\ 11)(5\ 13\ 19)(15\ 17\ 21), \\ b_3 &= (1\ 8\ 6)(2\ 22\ 19)(3\ 11\ 12)(5\ 14\ 20)(7\ 13\ 21)(15\ 23\ 16).\end{aligned}$$

For each of g_2 and g_3 , the exact class search finds 138 ordered factorizations of type $3A \cdot 3A$ and zero of types $2A \cdot 2A$, $2A \cdot 3A$, and $3A \cdot 2A$.

Computational proof. The refinement verifier reconstructs the $2A$ and $3A$ conjugacy classes in the released degree-23 copy of M_{23} , of sizes 3,795 and 56,672. For each target g_i , it solves $ab = g_i$ by iterating the first class and testing the forced second factor. It obtains the stated factorization counts, verifies the printed identities $a_2b_2 = g_2$ and $a_3b_3 = g_3$, and checks product one for the refined tuple. The generated subgroup has order 10,200,960. $\square$

Corollary 7.4 (Factorization obstruction for the five-branch class). *Every refinement from $(2A, 23A, 23B)$ to $(2A, 2A, 2A, 2A, 3A)$ using exactly two local monodromies over each 23-branch has factorization count zero.*

Proof. Beyond the original $2A$ entry, the two splits would have to use three further $2A$ entries and one $3A$ entry. Partitioning these four factors into two pairs forces one pair of type $2A \cdot 2A$ and the other of mixed type. The exact factorization count in theorem 7.3 is zero for all three such ordered types; alternatively, the involution-pair case is covered directly by theorem 7.2. $\square$

Theorem 7.5 (Class-restricted minimal six-branch common refinement). *A common six-branch Hurwitz component joins the two five-branch components and has class multiset*

$$\boxed{\{2A, 2A, 2A, 3A, 3A, 3A\}}.$$

In the left-action convention used for the released tuple, one product-one generating representative is

$$\begin{aligned} u &= (1\ 15\ 6)(2\ 18\ 16)(4\ 20\ 9)(5\ 8\ 22)(7\ 12\ 21)(10\ 14\ 19), \\ v &= (1\ 6\ 16)(2\ 15\ 18)(4\ 7\ 21)(8\ 22\ 13)(9\ 20\ 12)(10\ 17\ 19), \\ h_3 &= (1\ 15)(2\ 19)(3\ 8)(4\ 17)(6\ 7)(11\ 23)(12\ 21)(14\ 18), \\ h_4 &= (2\ 7)(4\ 13)(6\ 8)(9\ 22)(10\ 11)(14\ 17)(15\ 20)(16\ 18), \\ h_5 &= (1\ 20)(3\ 7)(4\ 18)(6\ 19)(9\ 22)(10\ 23)(13\ 17)(14\ 16), \\ h_6 &= (3\ 19\ 14)(4\ 21\ 15)(5\ 7\ 8)(10\ 17\ 11)(12\ 16\ 20)(13\ 22\ 18). \end{aligned}$$

The tuple $(u, v, h_3, h_4, h_5, h_6)$ has class type $(3A, 3A, 2A, 2A, 2A, 3A)$ and generates M_{23} . It has two certified collision faces:

- (a) *fusing uv recovers exactly the printed seed of the $(2A, 2A, 2A, 2A, 3A)$ component;*
- (b) *fusing $h_4h_5 \in 3A$ and reordering gives a tuple of type $(2A, 3A, 3A, 3A, 3A)$ lying in the same colored braid component as the five-branch refinement of the released $(2A, 23A, 23B)$ tuple.*

Among common refinements with six local classes in $\{2A, 3A\}$ and one elementary fusion producing each five-branch face, six branches are minimal and the two target passports force the class multiset.

Exact finite proof. The bridge certificate verifies product one, class membership, and generated group order 10,200,960. It checks $uv = h_1$, where h_1 is the first $2A$ entry of the verified five-branch seed; the complete $3A$ class search gives 2688 ordered $3A \cdot 3A$ factorizations of this $2A$ element. Thus, collision (a) recovers the five-branch seed exactly.

For the other face, the certificate verifies $h_4h_5 \in 3A$. After the indicated collision and two Hurwitz moves that place the unique $2A$ entry first, one obtains an ordered tuple of type $(2A, 3A, 3A, 3A, 3A)$. The two reordering moves are q_2q_1 . An explicit word of length 14 in the colored generators q_1^2, q_2, q_3, q_4 and their inverses then carries the five-branch refinement of theorem 7.3 to the resulting inner Nielsen class. Because two letters are q_1^2 , the word has length 16 in the ordinary braid generators. The certificate bundle stores the word, its SHA-256 digest, and an independent verifier that reapplies every move before final canonicalization. This colored five-branch certificate is independent of the 29,005-move full B_{10} certificate of theorem 7.8.

For the stated class-restricted problem, minimality is formal. The two target passports are distinct and both have five branches, so a strict common refinement by one elementary split requires at least six branches. At six branches, obtaining $(2A, 3A^4)$ by one fusion and $(2A^4, 3A)$ by another forces one $2A, 2A$ pair to fuse to $3A$ and one $3A, 3A$ pair to fuse to $2A$. The source multiset is therefore $(2A^3, 3A^3)$, which the displayed tuple realizes. $\square$

Theorem 7.6 (Algebraic models of the two collision tails). *For the $3A + 3A \rightarrow 2A$ collision $uv = h_1$ of theorem 7.5, the degree-23 tail decomposes over $\mathbb{Q}$ as*

$$2\alpha_6 \sqcup 2\alpha_4 \sqcup 3\text{id},$$

where

$$\alpha_6(x) = \left(\frac{12x^2 + 12x - 1}{12x^2 - 12x - 1} \right)^3, \quad \alpha_4(x) = \frac{4x(x-2)^3}{(2x^2 + 10x - 1)^2}.$$

With branch values ordered as $(0, 1, \infty)$, their passports are

$$\alpha_6 : \quad (3, 3), (2, 2, 1, 1), (3, 3),$$

and

$$\alpha_4 : \quad (3, 1), (3, 1), (2, 2).$$

For the $2A + 2A \rightarrow 3A$ collision $h_4 h_5 \in 3A$, the degree-23 tail decomposes over $\mathbb{Q}$ as

$$\beta_6 \sqcup 4\beta_3 \sqcup \beta_2 \sqcup 3\text{id},$$

where

$$\beta_6(x) = \frac{(x^3 + 1)^2}{4x^3}, \quad \beta_3(x) = \frac{4x^3 - 3x + 1}{2}, \quad \beta_2(x) = \frac{x^2}{x^2 + 1}.$$

Their passports are

$$\beta_6 : \quad (2, 2, 2), (2, 2, 2), (3, 3),$$

$$\beta_3 : \quad (2, 1), (2, 1), (3), \quad \beta_2 : \quad (2), (2), (1, 1).$$

For every nontrivial component above, the corresponding transitive small-degree Nielsen triple is unique up to simultaneous conjugacy. Hence, these rational maps realize the actual restricted permutation triples of the two collision tails.

Exact algebraic and finite proof. For the first face, the subgroup $\langle u, v \rangle$ has natural-orbit degrees

$$6, 6, 4, 4, 1, 1, 1.$$

The identities

$$(12x^2 + 12x - 1)^3 - (12x^2 - 12x - 1)^3 = 72x(12x^2 + 1)^2$$

and

$$4x(x-2)^3 - (2x^2 + 10x - 1)^2 = -(4x+1)^3$$

give the stated branch multiplicities directly. For the second face, $\langle h_4, h_5 \rangle$ has natural-orbit degrees

$$6, 3, 3, 3, 3, 2, 1, 1, 1,$$

and direct factorization of $\beta_6, \beta_3, \beta_2$ and of their differences from 1 gives the displayed passports. The collision-tail certificate independently restricts the printed degree-23 permutations to every natural orbit and compares the resulting cycle profiles. Exhaustive enumeration in S_2, S_3, S_4, S_6 gives one simultaneous-conjugacy orbit for each required transitive passport, so profile agreement identifies the restricted triples. $\square$

Corollary 7.7 (Arithmetic genus check for the common refinement). *The two admissible collision presentations have arithmetic genus 8, equal to the genus of a smooth degree-23 cover with class multiset $(2A^3, 3A^3)$.*

Proof. For the six-branch passport,

$$2g - 2 = -46 + 3 \cdot 8 + 3 \cdot 12 = 14,$$

so $g = 8$. On the $3A + 3A \rightarrow 2A$ face, the tail has seven connected components and the $2A$ node has 15 branches; adjoining the genus-zero main component gives first Betti number $15 - (7 + 1) + 1 = 8$. On the $2A + 2A \rightarrow 3A$ face, the tail has nine components and the $3A$ node has 11 branches, so the dual graph has first Betti number $11 - (9 + 1) + 1 = 2$. The main $(2A, 3A^4)$ component has genus 6 by Riemann–Hurwitz, giving $6 + 2 = 8$. $\square$

7.1 A ten-involution refinement layer

Theorem 7.8 (Exact ten-involution refinements and five-pair fusion). *The following exact statements hold.*

- (a) *The printed five-branch seed of type $(2A, 2A, 2A, 2A, 3A)$ admits an explicit product-one generating refinement*

$$T = (t_1, \dots, t_{10}) \in (2A)^{10}$$

whose five consecutive pair products recover that seed. The associated degree-23 source genus is 18.

- (b) *An explicit word of length 29,005 in the full braid generators $q_1, \dots, q_9$ carries T to a ten-tuple whose consecutive $2 + 4 + 4$ block products are simultaneously conjugate to the released tuple (g_1, g_2, g_3) of type $(2A, 23A, 23B)$.*

- (c) *In the released degree-23 copy there is a unique element*

$$h = (1\ 5)(2\ 22)(3\ 8)(7\ 20)(10\ 17)(11\ 21)(12\ 14)(18\ 23)$$

in $C_{M_{23}}(g_1)$ satisfying

$$hg_2^{-1}h^{-1} = g_3, \quad hg_3^{-1}h^{-1} = g_2.$$

It is an involution in class $2A$.

- (d) *Independently of the ten-tuple in (b), the released tuple admits a product-one generating refinement*

$$R = (r_1, \dots, r_{10}) \in (2A)^{10}$$

with exact quadratic branch-cycle symmetry

$$\sigma_h(r_i) = r_{\iota(i)}, \quad \sigma_h(x) = hx^{-1}h^{-1}, \quad \iota = (3\ 10)(4\ 9)(5\ 8)(6\ 7),$$

with positions 1 and 2 fixed.

- (e) *A twelve-move interleaving braid sends R to a ten-tuple U whose consecutive pair products*

$$P_i = u_{2i-1}u_{2i} \quad (1 \leq i \leq 5)$$

have orders $(2, 2, 3, 2, 2)$, product one, and generated group M_{23} . An eight-move full- B_5 word carries the printed target five-branch seed to $P = (P_1, \dots, P_5)$ modulo simultaneous conjugacy. Moreover, $\text{ord}(P_3P_4) = 11$.

Exact computational proof. The ten-involution verifier reconstructs both printed ten-tuples, replays every letter of the 29,005-move and twelve-move words, and verifies all block products. The first tuple has ten entries of index eight, so

$$2g - 2 = -46 + 10 \cdot 8 = 34,$$

which gives $g = 18$. Centralizer enumeration proves uniqueness and the displayed identities for h . The verifier checks the label involution entry by entry. The five fused entries have the stated orders and product, and their generated subgroup has order 10,200,960. The verifier replays the stored eight-move word in the canonical five-branch quotient and reaches the canonical key of P . $\square$

Theorem 7.9 (Framed multiplier stabilization packet). *Let h be the descent involution of theorem 7.8. Applying*

$$q_3^{-1} q_2^{-1} q_1^2 q_2^{-1} q_1 q_3^2 q_2^{-1}$$

to the released four-factor 23A block and defining the 23B block by the same h -equivariant reversal gives a product-one $2A^{10}$ refinement R^{st} . The released twelve-move interleaving fuses R^{st} to a generating tuple P^{st} of orders $(2, 2, 3, 2, 2)$, and

$$q_4 q_3^{-1} q_4 q_3^{-1} q_2 q_1^{-1} q_2 q_1 q_3$$

carries the printed target seed to P^{st} modulo simultaneous conjugacy.

For the four entries $A_1, \dots, A_4$ of the stabilized 23A block, $(\text{ord}(hA_1), \dots, \text{ord}(hA_4)) = (4, 3, 5, 3)$. With $a = A_3$ and $r = ha$, exactly three elements $x \in C_{M_{23}}(h)$ satisfy

$$x^2 = h, \quad x^{-1}rx = r^2,$$

and therefore $\langle r, h \rangle \cong D_{10} < \langle r, x \rangle \cong F_{20}$. The set $\mathcal{P} = \{r^i, r^i h : i \in \mathbb{Z}/5\mathbb{Z}\}$ has ten rigidified markings; right multiplication by h pairs them freely and conjugation by x induces multiplication by 2 on the five cyclic positions.

Moreover,

$$(\mathbb{Z}/30\mathbb{Z})^\times = \langle 7 \rangle \times \langle 11 \rangle, \quad 7 \equiv (2, 1), \quad 11 \equiv (1, -1) \pmod{(5, 3)}.$$

Thus, the action of x on the stabilized marking represents the order-five multiplier required by the framed arithmetic transport law, while exchanging the two involution factors u_5, u_6 of the unique 3A pair represents the independent order-three inversion, since $u_6 u_5 = (u_5 u_6)^{-1}$.

Exact finite and frame-theoretic proof. The certificate replays both nine-move words and the twelve-move interleaving, verifies h -equivariance, the target canonical key, and generated group order 10,200,960. It enumerates $C_{M_{23}}(h)$ of order 2688, finds exactly three stated multiplier lifts, and verifies $|\langle r, x \rangle| = 20$. The unit-group decomposition and the factor-swap identity are direct. In the ordered pointed Hurwitz frame, arithmetic transport uses one cyclotomic exponent and one common frame conjugator; forgetting that frame is a separate quotient, and theorem 7.13 shows that the raw dual representatives then become one inner Nielsen point. The theorem identifies the required finite multiplier model on the rigidified packet; realization of a specified element of $G_{\mathbb{Q}}$ by a chosen x is a separate arithmetic datum. $\square$

Remark 7.10 (Ordering, the elliptic fiber, and continuation). The reduced inertia over $j = 1728$ is 2^{1470} , so every elliptic point has ramification index two. The construction keeps the equal-class ordering and fiber frame through arithmetic transport and passes to the reduced quotient afterward. The marking packet $\mathcal{P}$ lives on this framed refinement. The iterated degree-ten carrier in theorem 6.13 expresses the same ordered continuation through an explicit infinite tower of finite refinement levels.

Proposition 7.11 (Local factorization orbit structure). *Fix $z \in 2A$. Under conjugation by $C_{M_{23}}(z)$, the 2688 ordered factorizations*

$$z = uv, \quad u, v \in 3A,$$

split into four orbits of sizes

$$\boxed{1344, 896, 224, 224.}$$

The orbit used in theorem 7.5 is the unique orbit whose subgroup $\langle u, v \rangle$ has natural-sheet orbits

$$6 + 6 + 4 + 4 + 1 + 1 + 1;$$

it has size 1344 and factorization stabilizer of order two.

The 98 ordered factorizations

$$z = ab, \quad a, b \in 2A,$$

split into two centralizer orbits of sizes

$$\boxed{84 + 14.}$$

Their natural-sheet orbit decompositions are respectively

$$4 + 4 + 2 + 2 + 2 + 2 + 2 + 2 + 1 + 1 + 1$$

and

$$4 + 4 + 4 + 4 + 1 + 1 + 1 + 1 + 1 + 1 + 1.$$

Exact computational proof. Two independent centralizer computations enumerate the complete $3A$ and $2A$ classes, force the second factor from the first, enumerate the relevant centralizers, and close every conjugation orbit. The first computation returns 2688 factorizations and orbit sizes 1344, 896, 224, 224; the natural-orbit census singles out the displayed 1344 orbit. The second returns 98 factorizations and orbit sizes 84, 14 with the two displayed sheet decompositions. Since ab is an involution exactly when the involutions a and b commute, every nontrivial transitive local group in the second calculation is a quotient of the Klein four group. $\square$

Proposition 7.12 (Rational local models and the five-pair genus ledger). *Put*

$$\nu_4(x) = \frac{(x^2 + 1)^2}{4x^2}, \quad \nu_2(x) = \frac{x^2}{x^2 + 1}.$$

The map ν_4 has passport

$$(2, 2), (2, 2), (2, 2),$$

and ν_2 has passport

$$(2), (2), (1, 1)$$

up to permutation of the three branch values. Consequently, the size-84 $2A \cdot 2A \rightarrow 2A$ orbit has local tail model

$$2\nu_4 \sqcup 6\nu_2^{(\pi)} \sqcup 3\text{id},$$

where the six degree-two components are appropriate S_3 -translates of ν_2 , while the size-14 orbit has model

$$4\nu_4 \sqcup 7\text{id}.$$

For the ten-tuple U of theorem 7.8, pairs 1 and 5 have the size-84 profile, pairs 2 and 4 have the size-14 profile, and pair 3 has the $2A \cdot 2A \rightarrow 3A$ tail

$$\beta_6 \sqcup 4\beta_3 \sqcup \beta_2 \sqcup 3\text{id}$$

of theorem 7.6. Their arithmetic-genus contributions are

$$\boxed{4, 4, 2, 4, 4,}$$

whose sum is 18.

Proof. The identities

$$(x^2 + 1)^2 - 4x^2 = (x^2 - 1)^2$$

and the pole orders at 0 and ∞ give the passport of ν_4 . The fibers of ν_2 over $0, 1, \infty$ have profiles $(2), (2), (1, 1)$. Restricting the five printed involution pairs to their natural-sheet orbits gives exactly the decompositions stated above.

For a $2A$ node, there are 15 branches. Both $2A \cdot 2A \rightarrow 2A$ tail types have 11 connected components, so each contributes $15 - 11 = 4$ to the first Betti number of the dual graph. The $3A$ node has 11 branches and its tail has nine components, contributing $11 - 9 = 2$. Thus, the five simultaneous tails contribute $4 + 4 + 2 + 4 + 4 = 18$, equal to the source genus of $2A^{10}$. $\square$

Proposition 7.13 (Conjugacy collapse of the raw dual tuple). *Let*

$$R = (r_1, \dots, r_{10})$$

be the branch-cycle-equivariant refinement of theorem 7.8, and define

$$\sigma_h(x) = hx^{-1}h^{-1}, \quad R^\vee = (\sigma_h(r_1), \dots, \sigma_h(r_{10})).$$

Every r_i belongs to $2A$ and is therefore an involution. Consequently,

$$\boxed{R^\vee = hRh^{-1}}$$

entry by entry. Apply the same twelve-move interleaving braid to R and $R^\vee$ to obtain U and $U^\vee$, and let $P, P^\vee$ denote their five consecutive pair products. Then

$$\boxed{U^\vee = hUh^{-1}, \quad P^\vee = hPh^{-1}.}$$

Thus,

$$[R^\vee] = [R], \quad [U^\vee] = [U], \quad [P^\vee] = [P]$$

in the corresponding inner Nielsen moduli. Thus, the two permutation representatives determine one inner Nielsen point, and their representative-level exchange remains inside that point. The exact identities

$$h^2 = 1, \quad \sigma_h(h) = h, \quad h\sigma_h(h) = 1$$

give the representative-level symmetry and cocycle identities.

Proof. Since $r_i^2 = 1$,

$$\sigma_h(r_i) = hr_i^{-1}h^{-1} = hr_ih^{-1}$$

for every i , proving the first identity. Hurwitz moves commute with simultaneous conjugation, so applying the same braid word to both tuples preserves this equality and gives $U^\vee = hUh^{-1}$. Pair multiplication then gives $P_i^\vee = hP_ih^{-1}$ for every i . Inner Nielsen classes are simultaneous-conjugacy classes, which proves the moduli statement. The exact permutation certificate checks the last three identities directly. $\square$

Proposition 7.14 (Exact fixed/dual fiber of the cyclic quotient). *Put*

$$\beta_5(t) = 1 - (1 - t)^5, \quad x = 1 - t,$$

and let

$$Z_5 = \beta_5^{-1}(0) = \text{Spec } A_5, \quad A_5 = \mathbb{Q}[x]/(x^5 - 1).$$

Then Z_5 is finite étale of degree five over $\mathbb{Q}$ and

$$\boxed{A_5 \simeq \mathbb{Q} \times \mathbb{Q}(\zeta_5).}$$

The involution

$$j(x) = x^{-1} = x^4$$

has coefficients in $\mathbb{Q}$; in the t -coordinate it is

$$j(t) = \frac{t}{t-1}.$$

Its scheme-theoretic fixed locus is exactly

$$Z_5^j = \{x = 1\} = \{t = 0\} \simeq \operatorname{Spec} \mathbb{Q}.$$

The primitive central idempotent of this fixed factor is

$$e_{\text{fix}} = \frac{1 + x + x^2 + x^3 + x^4}{5},$$

and multiplication by e_{fix} on A_5 has rank one and trace one.

Proof. The factorization

$$x^5 - 1 = (x - 1)\Phi_5(x)$$

is square-free in characteristic zero, and Φ_5 is irreducible over $\mathbb{Q}$. The Chinese remainder theorem therefore gives

$$A_5 \simeq \mathbb{Q}[x]/(x - 1) \times \mathbb{Q}[x]/(\Phi_5) \simeq \mathbb{Q} \times \mathbb{Q}(\zeta_5).$$

On A_5 , inversion is $x \mapsto x^4$. The fixed subscheme is cut out by $x - x^4$ together with $x^5 - 1$, and

$$\gcd(x^5 - 1, x - x^4) = x - 1.$$

Hence, the fixed locus is $\operatorname{Spec} \mathbb{Q}$. The displayed idempotent evaluates to one at $x = 1$ and to zero at every nontrivial fifth root of unity, so it is the central projector of the $\mathbb{Q}$ -factor. Its multiplication matrix has rank one and trace one. $\square$

Proposition 7.15 (Quadratic dual complement and ledger identification). *Let j be the involution of theorem 7.14 and put*

$$y = x + x^{-1}.$$

Then

$$A_5^j \simeq \mathbb{Q}[y]/((y - 2)(y^2 + y - 1)) \simeq \mathbb{Q} \times \mathbb{Q}(\sqrt{5}).$$

On the nonfixed factor,

$$B = \mathbb{Q}[x]/(\Phi_5(x)) \simeq \mathbb{Q}(\zeta_5)$$

one has

$$B^j = \mathbb{Q}(\sqrt{5}), \quad [B : B^j] = 2.$$

Thus,

$$\operatorname{Spec} \mathbb{Q}(\zeta_5) \longrightarrow \operatorname{Spec} \mathbb{Q}(\sqrt{5})$$

is the genuine finite étale quadratic dual quotient in the cyclic fiber.

Set $K = \mathbb{Q}(\sqrt{5})$. After base change to K , define

$$N = \frac{\sqrt{5}}{5} (-y^2 + y + 2) \in A_5^j \otimes_{\mathbb{Q}} K.$$

Then

$$N^3 = N,$$

and N takes the values

$$0, \quad +1, \quad -1$$

on the three j -orbits

$$\{1\}, \quad \{\zeta_5, \zeta_5^{-1}\}, \quad \{\zeta_5^2, \zeta_5^{-2}\}.$$

Consequently,

$$A_5^j \otimes_{\mathbb{Q}} K \simeq K[N]/(N^3 - N),$$

with projectors

$$e_0 = 1 - N^2, \quad e_+ = \frac{N^2 + N}{2}, \quad e_- = \frac{N^2 - N}{2}.$$

Moreover,

$$e_0 = e_{\text{fix}} = \frac{y^2 + y - 1}{5},$$

while the nontrivial element of $\text{Gal}(K/\mathbb{Q})$ sends

$$N \mapsto -N, \quad e_+ \longleftrightarrow e_-, \quad e_0 \mapsto e_0.$$

Finally,

$$\beta_5(e_0) = e_0, \quad \beta_5(1 - N^2) = 1 - N^{10} = e_0.$$

Thus, the fixed point, the two dual outcomes, the exact 2:1 quotient, and the trivalent ledger describe the same finite étale decomposition after base change to the quadratic splitting field K .

Proof. The invariant subspace of A_5 under $x \mapsto x^{-1}$ has basis

$$1, \quad x + x^{-1}, \quad x^2 + x^{-2}.$$

Since

$$y^2 = 2 + x^2 + x^{-2},$$

the element y generates the invariant algebra. Direct reduction modulo $x^5 - 1$ gives

$$(y - 2)(y^2 + y - 1) = 0,$$

which proves the invariant-algebra presentation. The quadratic factor has discriminant 5, giving the second displayed isomorphism.

On $B = \mathbb{Q}(\zeta_5)$, inversion is complex conjugation, so its fixed field is the maximal real subfield

$$\mathbb{Q}(\zeta_5 + \zeta_5^{-1}) = \mathbb{Q}(\sqrt{5}).$$

The extension therefore has degree two.

Over K , the three roots of the invariant polynomial are

$$2, \quad \frac{\sqrt{5} - 1}{2}, \quad -\frac{\sqrt{5} + 1}{2}.$$

Substitution into the displayed formula for N gives $0, 1, -1$. Therefore, $N^3 = N$ and N separates the three factors, proving the ledger isomorphism. Reduction modulo $(y - 2)(y^2 + y - 1)$ gives

$$1 - N^2 = \frac{y^2 + y - 1}{5},$$

which is exactly the projector of the fixed factor. Conjugating $\sqrt{5}$ changes N to $-N$ and exchanges the two nonfixed projectors. The last identities follow from idempotency of e_0 and from $N^3 = N$. $\square$

Corollary 7.16 (Fixed/dual decomposition of the cyclic fiber). *Central idempotents split $Z_5 = \beta_5^{-1}(0)$ as*

$$Z_5 = \operatorname{Spec} \mathbb{Q} \sqcup \operatorname{Spec} \mathbb{Q}(\zeta_5), \quad Z_5^J = \operatorname{Spec} \mathbb{Q}.$$

The primitive central trace-one projector e_{fix} selects the $\mathbb{Q}$ -rational fixed summand. On the complementary factor, inversion has fixed field $\mathbb{Q}(\sqrt{5})$ and gives the finite étale quadratic extension

$$\mathbb{Q}(\zeta_5)/\mathbb{Q}(\sqrt{5}).$$

Definition 7.17 (Rigidified pair-marked boundary stack). *Let $\mathcal{H}_5^{\text{ord}}$ denote the full ordered Hurwitz stack over $\mathbb{Q}$ of connected degree-23 covers of type $(2A, 2A, 3A, 2A, 2A)$. Let*

$$\tilde{\mathcal{H}}_5^{\text{pair}} \longrightarrow \mathcal{H}_5^{\text{ord}}$$

be the finite marking stack which records, for each of the five local monodromies, an ordered $2A \cdot 2A$ factorization of the same local type as the corresponding pair of U , together with rigidifications identifying the associated tail with the printed rational tail model.

Let $\mathfrak{D}_{\text{pair}}$ be the corresponding balanced pair-clutching boundary stratum. In the standard product description of a fully marked admissible-cover boundary stratum, one factor is the central five-point cover and the other five factors are the fixed rational tail covers of theorem 7.12; the factors are glued along the specified node-evaluation sections.

Theorem 7.18 (Global pair-clutching and core stabilization). *There are morphisms of algebraic stacks over $\mathbb{Q}$*

$$\tilde{\mathcal{H}}_5^{\text{pair}} \xrightarrow{J} \mathfrak{D}_{\text{pair}} \xrightarrow{S} \mathcal{H}_5^{\text{ord}}$$

such that

$$S \circ J = \pi,$$

where π forgets the pair markings. The morphism J attaches the five fixed rational tail covers, while S takes the normalized inverse image of the central target component and regards the five nodes as branch sections with monodromy equal to the corresponding pair product.

Proof. The explicit tail maps of theorem 7.12 lie over $\mathbb{Q}$, and their node passports agree with the restrictions of the five printed permutation pairs. Standard admissible-cover clutching therefore defines J after the stated finite marking and rigidification. By construction of the marked boundary stratum, projection to the central factor defines S ; forgetting the five node markings turns that factor into the fused five-branch cover. Attaching the five fixed tails and then projecting back to the central factor recovers the original cover after the pair markings are forgotten, so $S \circ J = \pi$. This is the standard clutching/product description of marked admissible-cover boundary strata, expressed on the balanced twisted-cover normalization [5, 10]. $\square$

Theorem 7.19 (Balanced five-node smoothing). *Let ξ be a geometric point of $\mathfrak{D}_{\text{pair}}$. After passage to an étale neighborhood of ξ , each of the five balanced target nodes has a local chart*

$$[\operatorname{Spec} A[z_i, w_i]/(z_i w_i - t_i) / \mu_{r_i}], \quad (z_i, w_i) \longmapsto (\zeta z_i, \zeta^{-1} w_i).$$

The five disjoint nodes admit simultaneous local smoothing parameters $t_1, \dots, t_5$. The diagonal specialization

$$t_1 = \dots = t_5 = t$$

defines an étale-local algebraic one-parameter smoothing of all five nodes over the residue field of the pair-marked admissible cover.

Proof. For a twisted node, Abramovich–Corti–Vistoli give an étale-local quotient chart of the displayed form, with inverse tangent characters in the balanced case, and identify balanced twisted covers with the normalization framework for admissible covers. For a marked nodal target, disjoint nodes contribute independent local smoothing parameters; pulling those parameters back to the balanced cover chart gives $t_1, \dots, t_5$. Restriction to the diagonal $t_1 = \dots = t_5$ gives the stated étale-local algebraic one-parameter smoothing [5]. $\square$

Corollary 7.20 (Closed geometric mobility diagram). *At the Nielsen-class and admissible-boundary level, the explicit connected refinement network is*

$$(2A, 23A, 23B) \longleftarrow (2A, 3A^4) \longleftarrow (2A^3, 3A^3) \longrightarrow (2A^4, 3A) \longrightarrow (2A, 2A, 3A, 5A),$$

where exact collision certificates give the first, middle, and last arrows, and the braid word above certifies membership of the middle-left face in the released five-branch refinement component. At the all-involution level, the exact certificates give

$$(2A, 23A, 23B) \longleftarrow 2A^{10} \longrightarrow (2A^4, 3A),$$

in the precise two-part sense of theorem 7.8: one B_{10} component contains refinements of the released triple and the target seed, while a separately certified branch-cycle-equivariant refinement fuses into the verified target five-branch component.

8 Computational certificates

The finite computations and the geometric deductions are separated as follows. “Machine” means that the stated finite or symbolic datum is recomputed exactly by the released verifier. “Machine + prose” means that the finite input is machine-verified and the geometric conclusion is proved in the cited argument of the article.

Main theorem	Verification layer	Public certificate or proof layer
(i), (iii)	Machine	<code>computations/core/geometry_certificates/verify_m23_ordered_hurwitz.py</code> , <code>computations/core/geometry_certificates/verify_m23_nielsen_exhaustiveness.py</code> , and <code>computations/core/geometry_certificates/verify_m23_reduced_jline.py</code> .
(ii)	Machine + prose	<code>computations/arithmetic/code/verify_m23_real_boundary.py</code> certifies the finite boundary data; theorems 2.2 and 2.5 prove the $\mathbb{Q}$ -model and divisor consequences.
(iv)	Machine + prose	<code>computations/core/geometry_certificates/m23_five_branch_clutching.cpp</code> certifies the width-five cycles and clutching bijections; admissible-cover identification uses [6, 5, 10].
(v)	Machine + prose	The four- and five-branch equation builders and the B_{27} result files certify the algebraic data; section 3 states the source-activation criterion.
(vi)–(viii)	Machine + prose	<code>computations/core/m23_six_bridge_certificate.py</code> and <code>computations/core/m23_collision_tail_certificate.py</code> certify the finite words, factorizations, and passports; the article proves the corresponding clutching interpretation.
(ix)	Machine + prose	<code>computations/core/m23_recursive_pullback_certificate.py</code> certifies the set-stabilizer data; theorems 6.7 to 6.9 prove the base change, arithmetic descent, and quotient descent.

Main theorem	Verification layer	Public certificate or proof layer
(x)–(xii)	Machine + prose	<code>computations/core/m23_arithmetic_stabilization_certificate.py</code> and <code>computations/core/m23_five_pair_tail_certificate.py</code> certify the refinement, multiplier, factorization, and tail data; the associated fusion and stabilization are proved in the text.
(xiii)–(xv)	Machine + prose	<code>computations/core/m23_fixed_dual_fiber_certificate.py</code> , <code>computations/core/m23_dual_two_fives_certificate.cpp</code> , <code>computations/core/m23_dual_two_fives_projector.py</code> , and <code>computations/core/m23_ten_width11_completion_certificate.cpp</code> certify the finite-algebra, ten-state, native-return, and width-11 data; the article supplies the quotient and admissible-cover interpretations.

Every listed source, recorded output, and auxiliary datum is bound by the release-level `SHA256SUMS.txt`; the master replay compares fresh mathematical JSON content against the recorded outputs before reporting success. An independent GAP/CTbLib Frobenius cross-check is supplied as `computations/INDEPENDENT_GAP_CROSSCHECK.g`; when GAP with CTbLib is available, `RUN_ALL_CERTIFICATES.sh` verifies the counts 98, 2688, and 127,200 independently of the Python/C++ braid computations. The accompanying release contains exact source verifiers, stored outputs, coefficient files, and SHA-256 hashes for every finite computation used above. The folder `computations/core/` reconstructs the 980-state ordered class, exhausts the 14,402,025 fixed-5A pair search, reconstructs the 11,760 all-orderings orbit and degree-2940 reduced j -line, enumerates the complete 21,456 five-branch component and its clutching boundaries, expands the 48- and 72-equation systems, and verifies the six- and ten-branch refinement certificates.

The arithmetic stabilization source `computations/core/m23_arithmetic_stabilization_certificate.py` verifies the h -equivariant refinement, target-component word, $D_{10} < F_{20}$ marking packet, multiplier-2 lifts, and order-3 pair-swap inversion. The source `computations/core/m23_dual_two_fives_certificate.cpp` reconstructs the selected ten-state cell and canonical five phase pairs; `computations/core/m23_dual_two_fives_projector.py` verifies the rank-five projectors, multiplier action, pairing quotient, and β_{10} model. The source `computations/core/m23_ten_width11_completion_certificate.cpp` reconstructs the same 21,456-state component and verifies the native return $\kappa_{\text{ret}} q_1 \kappa_{\text{ret}} = q_1^{-1}$, the dihedral order-10 action, all 264 width-11 cusps, the four degree-66 genus-zero components, and the component-pairing involution κ_{11} . The supplied subfolder `computations/core/width11_boundary_resolution/` preserves the independent width-11 verifier and recorded output reproducing the four degree-66 passports, genus calculations, and inverse-return component pairing. The source `computations/core/m23_recursive_pullback_certificate.py` verifies the order-92,897,280 mixed-boundary group, the central deck involution, the 126-subset orbit, the transitive ten-state stabilizer action, exact agreement with the canonical κ_{pair} -pairing, the induced S_5 quotient, and the genera 31, 327, and 161. The source `computations/core/m23_comparison_morphism_certificate.py` verifies the finite hypotheses used by theorem 6.9: the global quotient source $\overline{Q}_H$, genus 161, the central-involution equivariance data, finite quotient descent, and the separate genus-zero $1^5, 5, 5$ cyclic carrier represented by β_5 . The source `computations/core/m23_raw_dickson_separation_certificate.cpp` verifies five 21,456-state orbits for the raw pure simultaneous-collision subgroup.

The folder `computations/arithmetic/` contains the real-boundary verifier, the degree-27 target-algebra checks, the reduced j -line arithmetic data, boundary and transport checks, and the universal coefficient-model data. The root script `RUN_ALL_CERTIFICATES.sh` dispatches the public verification suite with Python bytecode generation disabled. `STRUCTURE.md` records the package map, and `SHA256SUMS.txt` binds the public deposit.

Data and code availability. The complete computational release is assigned Zenodo DOI 10.5281/zenodo.22010546. The deposit contains the manuscript sources, deterministic verification programs, recorded machine-readable outputs, replay script, environment record, and SHA-256 integrity manifest.

9 Conclusion

The ordered $(2A, 2A, 3A, 5A)$ Nielsen class contains exactly 980 generating states and gives a genus-119 cross-ratio Hurwitz curve. The full all-orderings construction produces a degree-2940, genus-43 reduced j -line cover with 406 cusps. The colored $(2A^4, 3A)$ component contains 21,456 states, and all six equal- $2A$ width-five collision divisors return the complete 980-state four-branch component by admissible-cover clutching and equal-class braid symmetry.

The real-boundary verification gives eight individual $\mathbb{Q}$ -rational points on the compactified ordered curve, while the two width-one degree-two divisors over $\lambda = 1$ and $\lambda = \infty$ are imaginary quadratic. Interior specialization is recorded separately through the degree-27 target algebra $B_{27} = \mathbb{Q}[Z]/(Z^{27} - Z)$ and its trace-one projectors, with the genuine degree-980 source algebra and algebra map serving as the activation datum.

The explicit $(2A, 23A, 23B)$ triple admits a five-branch refinement of type $(2A, 3A^4)$. The class $(2A^3, 3A^3)$ forms a common six-branch refinement between that component and the $(2A^4, 3A)$ component. The rational maps $\alpha_6, \alpha_4, \beta_6, \beta_3, \beta_2$ realize the corresponding local admissible tails over $\mathbb{Q}$. At the all-involution level, a 29,005-move braid identifies a $2A^{10}$ refinement of the target seed with a refinement contracting to the explicit three-point triple, while a branch-cycle-equivariant ten-involution refinement fuses by five pairs into the verified target component. The local factorization orbit decompositions are $1344 + 896 + 224 + 224$ and $84 + 14$, and the five local genus contributions sum to the source genus 18.

The stabilization carries a certified $D_{10} < F_{20}$ marking: $x^2 = h$ realizes the order-five doubling action, while swapping the unique ordered $2A \cdot 2A \rightarrow 3A$ pair realizes inversion. The reduced $j = 1728$ computation is fixed-point-free of type 2^{1470} ; arithmetic stabilization therefore stays in the ordered frame before quotienting. The degree-ten carrier β_{10} iterates to finite covers of degree 10^n , giving the explicit infinite tower attached to this continuation.

The degree-five cyclic cover has zero-fiber algebra $A_5 = \mathbb{Q}[x]/(x^5 - 1) \simeq \mathbb{Q} \times \mathbb{Q}(\zeta_5)$. Inversion fixes exactly the $\mathbb{Q}$ -factor. The primitive central idempotent $e_{\text{fix}} = (1 + x + x^2 + x^3 + x^4)/5$ selects that factor and has multiplication rank and trace one. On the complementary cyclotomic factor, inversion has fixed field $K = \mathbb{Q}(\sqrt{5})$ and gives the degree-two étale extension $\mathbb{Q}(\zeta_5)/K$. After base change to K , the invariant algebra splits as $A_5^j \otimes_{\mathbb{Q}} K \simeq K[N]/(N^3 - N)$ with $e_0 = 1 - N^2$ and $e_{\pm} = (N^2 \pm N)/2$. The nontrivial element of $\text{Gal}(K/\mathbb{Q})$ fixes e_0 and exchanges e_+ with e_- . Thus, the rational fixed channel and the two conjugate split channels form one exact finite étale fixed/dual decomposition.

For the ten-involution symmetry, every refined entry is an involution, so $R^\vee = hRh^{-1}$, $U^\vee = hUh^{-1}$, and $P^\vee = hPh^{-1}$. These representatives therefore define one inner Nielsen point, while the representative-level cocycle identity $h\sigma_h(h) = 1$ remains exact. The rigidified pair-clutching morphism attaches the five explicit rational tails, central-core stabilization recovers the fused five-branch cover, and balanced twisted-cover deformation theory gives an étale-local simultaneous smoothing of the five nodes.

The mixed-boundary calculation geometrizes the ten-state cell after the canonical set-stabilizer base change and supplies the quotient comparison morphism $\mathfrak{B}_H$ by finite descent. The mixed-boundary monodromy group has order 92,897,280, the orbit of Δ_{10} has size 126, and the set stabilizer acts transitively on the ten states. The unique central deck involution restricts to the phase pairing κ_{pair} , producing the connected degree-10 pullback and its genus-161 degree-5 quotient. On the same ten states, the native return $\kappa_{\text{ret}}(n, p) = (1 - n, p + 1)$ satisfies $\kappa_{\text{ret}} q_1 \kappa_{\text{ret}} = q_1^{-1}$ and gives a dihedral group of order 10. The global width-11 mixed-boundary locus consists of 264 cusps in four degree-66, genus-zero components, paired by κ_{11} as $0 \leftrightarrow 2$ and $1 \leftrightarrow 3$. These results, together with the full six-direction width-five clutching and the six- and ten-branch refinement diagrams, give a single exact description of the positive-dimensional M_{23} Hurwitz geometry connected to the explicit three-point construction.

Acknowledgments

The authors congratulate Xiaoyu Huang, Blake Jackson, Kyu-Hwan Lee, Bjorn Poonen, Rachel Pries, and Shaowu Zhang on their resolution of the inverse Galois problem for M_{23} , and thank them for their encouragement to publish. OpenAI ChatGPT and Anthropic Claude were used for code generation, computational verification, error checking, and manuscript revision. All computational claims are tied to independently executable source files, recorded outputs, and cryptographic hashes in the accompanying release; the geometric deductions are proved in the text from those certified finite inputs.

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